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Enhancing Safety in Ultra-Sour Offshore Environments: A Comprehensive Emergency Response System

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Abstract

This paper presents the emergency response system implemented on the Hail and Ghasha Project during the execution phase. It integrates digital Personnel-On-Board accounts, real-time marine activity, and e-tracking for personnel monitoring working in an ultra-sour offshore environment. In combination with adequate training for the parties involved in the Drilling and Facilities construction Simultaneous Operations (SIMOPS), these components are crucial for the safety of personnel when a short response time is essential. The system's deployment on the artificial islands required for production of the Hail and Ghasha Offshore Fields demonstrates its effectiveness in managing complex challenges, enhancing safety protocols, and improving operational efficiency in high-risk offshore environments.

Introduction

The Hail and Ghasha (H&G) mega project is a development located in the Abu Dhabi offshore area within the Marwah Marine Protected Biosphere. Artificial islands have been constructed to serve as Drilling Centers (DCs) and to install the facilities required for the stabilization and transport of the ultra-sour production of the Ghasha and Hail fields to the onshore facilities for further processing. To achieve the program targets it is required that the Drilling and Construction of facilities is performed under Simultaneous Operations (SIMOPS).

This creates a particular situation where both Drilling Personnel and Facilities construction personnel are required to maintain a coordinated response in case of emergency. Of course, the most significant event under this condition is the potential for well blowout during drilling, which is a condition that cannot be discarded. Moreover, while constructing and drilling in this environment, the evacuation might become challenging for those workers that are required to work in or within the process equipment being built. Each island executes between 10–20 activities daily with several teams involved (e.g.: drilling crews, well testing teams, heavy construction, marine operations, etc.). The workforce includes approximately 140 personnel per drilling area and 300 for facilities construction per island, at peak construction up to 5000 workers could be affected in a fully escalated case affecting the whole field (e.g., well blowout, etc.).

In the case that a blowout condition takes place, the whole island could become immersed in a toxic H₂S cloud depending on factors such as wind direction and speed, direction of the blowout, capacity of

detection and action, etc. With the many variables involved, it is of paramount importance that the crews are trained and prepared to act immediately upon the notification of an emergency. Thus, there are three critical factors that define the success of the Emergency Response in these challenging conditions. First, rapid and coordinated communication. Second, well defined and practiced emergency plans. And third, availability and readiness of resources needed to produce a safe evacuation of all the personnel in the island.

This paper discusses the implementation of an organized Emergency Response, utilizing a coordinated approach for the safe evacuation of all personnel. To achieve this target, it is necessary to know the total number of personnel involved at each location and at any given time. Then, the organization should know exactly how to react in case a notification is given, and should be able to coordinate the rescue, evacuation, transportation, and communication efforts to all the stakeholders that could be affected. All these actions shall be performed in a timely manner.

The process applied in the Emergency Response Plan is following ADNOC's Corporate Practice for Emergency Response and Crisis Management, but also in the Incident Command System (ICS). In this way, the response can be escalated in a proven manner and follow a standardized approach as required. However, this paper is not an exhaustive description of the Emergency Response Plans, that include also equipment, personnel, agreements, documentation, and other aspects of the adequate response to different levels from incidents to crisis.

Statement of Theory and Definitions

ADNOC's Corporate Practice for Emergency Response and Crisis Management

This document presents the structural requirements to establish a comprehensive response to emergent situations. It also ensures that the system has appropriate capacity to tackle these situations and ensure Business Continuity. Organizational requirements are aligned with the Incident Management System (ICS), which is a standardized, on-scene, all-hazards incident management approach used globally. In this context, the words:

Level (1 to 3): Corresponds to the criticality level of response, where level 1 represents an immediate tactical response, while level 3 implies a more strategical response taken at higher levels of the organization, typically requiring external support coordination.

Tier (1 to 3): Corresponds to a given response to specifically address loss of containment. Where Tier 1 implies localized needs, while Tier 3 implies a broad response exceeding the local capacity of the project. Typically requiring external support coordination.

In case an event is triggered, the different levels of the organization are activated. Starting with the Tactical Response Teams, and the Incident Management Team.

Definitions

- **Tactical Response Team (TRT):** The first line responders with adequate training to overcome the incidents or emergencies.
- **On-Scene Commander (OC):** is the person in charge of the Tactical Response Team.
- **Incident Command Post (ICP):** The location where the TRT is based to cater for an emergency with adequately trained personnel.
- **Incident Management Team (IMT):** Organization that supports the TRT with logistic, finance, operations, planning, and HSE during an incident.
- **Incident Commander (IC):** The leader of the Incident Management Team.
- **Incident Command Center (ICC):** The place where the command-and-control facilities for the IMT is located.

- **Emergency Support Team (EST):** Team of people that supports the regional/project wise requirements for the needs of the IMT in the areas of logistics, finance, operations, planning and HSE during an Emergency.
- **Emergency Commander (EC):** Leader of the EST.
- **Emergency Support Center (ESS):** The place where the command-and-control facilities for the EST is located.
- **Crisis Management Team (CMT):** team of people that supports the overall response and ensures that the needs of the EST in the areas of logistics, finance, operations, planning and HSE during a Crisis.
- **Crisis Commander (CM):** is the person in charge of the Crisis Management Team. Typically, a Senior Manager of the Organization with the capacity to take strategic decisions, and coordinate support outside the organization.
- **Crisis Management Center (CMC):** The place where the command-and-control facilities for the EST is located.
- **Incident:** is an event, series of events or set of circumstances that interrupts normal operating procedures and has the potential to precipitate an emergency or crisis. (Ref. 1)
- **Emergency:** is an unforeseen combination of circumstances that does the following if not controlled, contained or eliminated immediately will Disrupts normal operating conditions or pose an actual or potential threat to people, environment and assets. (Ref. 1)
- **Crisis:** is an event, series of events or set of circumstances that threaten to fundamentally affect or alter the way ADNOC chooses to do business. (Ref. 1)
- **Dimensioning condition:** The identified condition upon will have as consequence the most significant effect requiring response class.
- **Response Class:** the classification of possible specialized response required (e.g.: Well Control, Fire and explosion, environmental impact, medical response, Rescue at site).

Description and Application of Equipment and Processes

Emergency Response Organization

The organization has three levels of response, each of them has a minimum structure that oversees the appropriate response at each Level/Tier. Thus, the response to any incident is a that of a tiered one. The principle being that there is always to command centres that coordinate action according to the needs of the situation.

As is possible to appreciate in the figure above, the combination of multiple major contractors serving under their own scope and responsibility during an event forces the separation of the TRTs and the ESTs. This is because not only is the Scope of Work different to each major Contractor, but also the nature of the incidents is different. For example, during the construction of facilities in a green field, there is minimal to no-risk associated with Loss of Containment that could feed an ever-escalating event. However, during the well drilling activity the dimensioning condition for the field is defined by the loss of well control, which potential could cascade in a full blowout condition. In the case of an ultra-sour reservoir this situation could have dire consequences for all the personnel operating in the artificial islands of Hail and Ghasha Project.

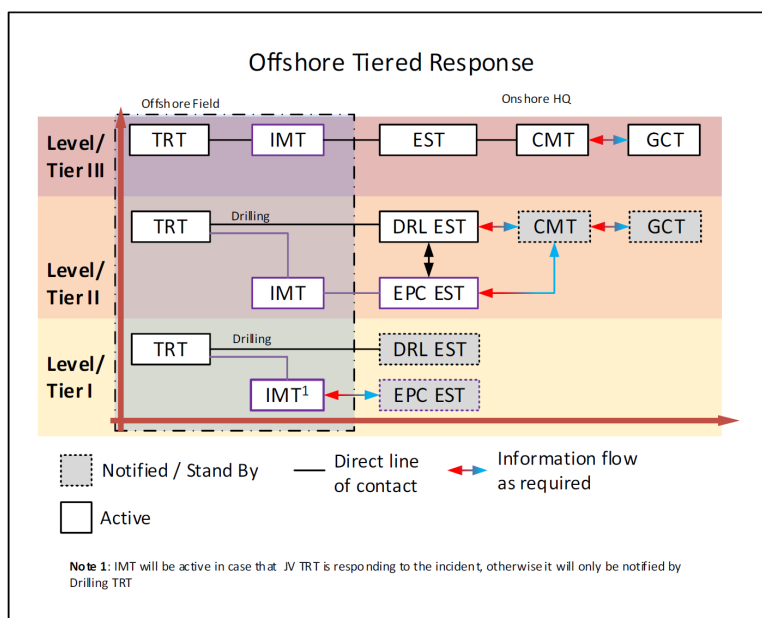


Figure 1—Tiered organization structure in Ghasha Concession Asset

Well Control Event

A well control event could be described in many ways, and based on well-known events in the industry, it can also be said that the events could occur in an unpredictable way. Some well control events are undetected until the hydrocarbon flow is on surface in a sequence of fast escalating events (e.g. Macondo;2010), while others are under the operational view but exceed the expectations or capacities of the drilling teams and could last days (e.g.: Ekofisk Bravo;1977). In either case, time is of essence, even in slow escalating events, time is usually misused or misunderstood. Thus, the need for an effective and organized evacuation is always the way forward in such cases.

According to the impact assessments, some well blowouts from ultra-sour reservoirs could produce up to a few kilometers of H_2S dispersion before concentrations fall under the accepted level of 99% survivable exposure or 76ppm as a value normally used for the definition of the Emergency Preparedness Area.

Therefore, it is imperative that the early detection of an abnormal condition is achieved. In this way the available time is utilized to produce a safe evacuation. When time is of essence, the possibility of the secondary events (e.g., falls and trips, lack of proper EBA usage, etc.) is increased due to the stress associated with taking fast decisions and actions (e.g. during the evacuation). This is particularly important for personnel that are new to the operation.

For the Hail and Ghasha project, the application solution is based on three pillars:

- First, utilization of available technology to continuously monitor the personnel at high-risk areas (i.e.: e-tracking system, and POB access control) and to ensure that other personnel at other locations are properly accounted for, for example, on boats during transportation or islands where the toxic exposure potential is limited.
- Second, is the coordination of marine operations with real-time position access of local and foreign vessels at location (i.e. Marine Tracking System). Since there are extensive zones where the toxic dispersion could be affected, it is important to be able to identify and notify to own operated vessels and external vessels in affected transit areas.
- Third, continuous training of personnel on the mustering and evacuation equipment and Toxic Gas Refuges.

In any case, the OC will alert the IC, who will in turn activate the IMT organization and inform the ECs of the situation. If the incident escalates to an emergency, the ECs will activate the ESTs and inform the CM. If the Emergency escalates to a crisis, the CM will activate the CMT and ensure notification to external structures of support.

Electronic Tracking System application

During the development of a blowout condition, two situations could arise:

1. Fast escalation, detected already with presence of H_2S in the immediate detectors at site.
2. Slow escalation, detected by Drilling operators during the activity. In this condition, early notification can be given to other parties in the SIMOPS condition at the Drilling Centers.

The following figure shows the disposition of a typical Drilling Center, where the activity of Drilling is taking place in parallel with the facilities construction, as seen by the e-tracking system.



Figure 2—Artificial Island with the electronic tracking system, and the clusters for outdoor usage

The system is capable to identify the user location, which in combination with cluster description allow the use to do an automatic count on the POB at each cluster. Thus, in case of emergency, when an evacuation is required, there is no need to do any sort of manual counting of personnel. The sole presence in the area will identify the worker, the position, and other critical information required from the individual in case of emergency. Similarly, if a person is not reaching the cluster destination in the case of emergency, the Tactical Response Team in charge of search and rescue will have an accurate target to pinpoint the efforts and reduce the exposure as much as possible.

As for the people accessing the muster areas (i.e.: the Toxic Gas Refugees) the counting is automatic, and the system has enough resolution as to give a specific accounting of the personnel reaching this ever important location in a condition of evacuation, as shown in Figure 3.

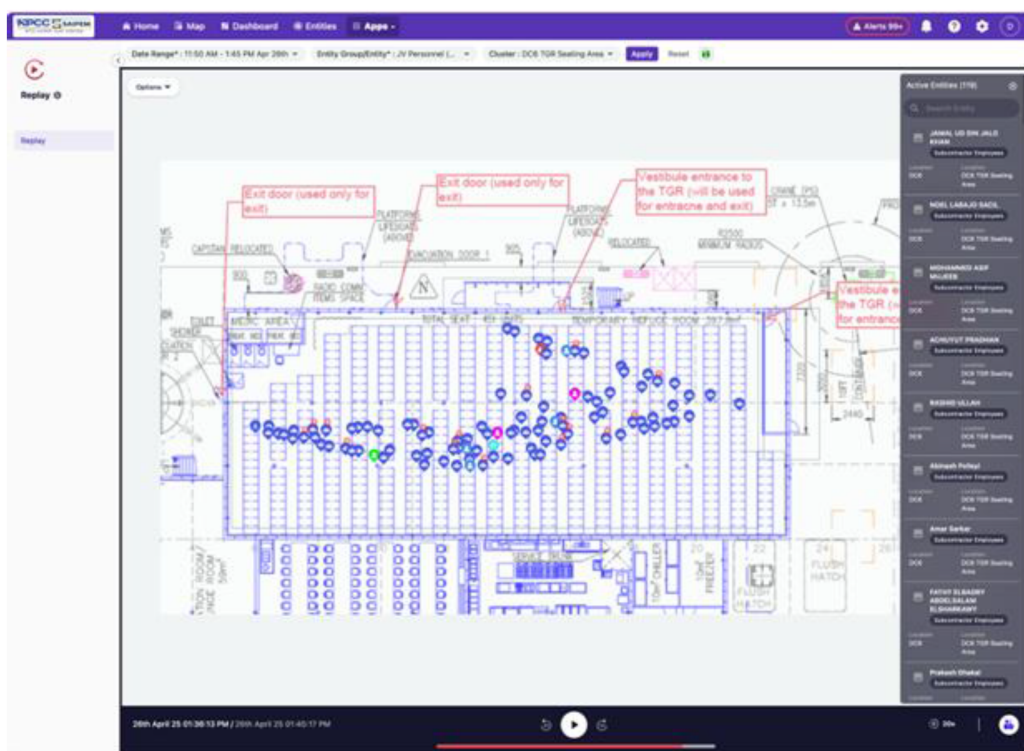


Figure 3—Muster station inside the accommodation barge with geo-localized personnel account

System Capability

The e-tracking system is a LoRa based application. LoRa stands for Long Range and is a wireless communication technology designed for low-power, long-range, and low-data-rate applications. With a very limited data pack to be sent, the battery of the device is long lasting, the size of a data pack to be transmitted every minute is about 2Kb. The architecture of the system was designed to ensure the scalability of the operation and resilient to the conditions of the operation. For example, in case of gas or toxic gas detection, the power generation is shutdown. However, the system is still operable as it is battery based, with repeaters at site that are solar powered. Additionally, in case of communications disconnection the gateway is connected to VSat data, allowing the transmission of information to the net via a satellite link. Thus, the primary and secondary electrical and communication feed give the system capability to perform at all times.

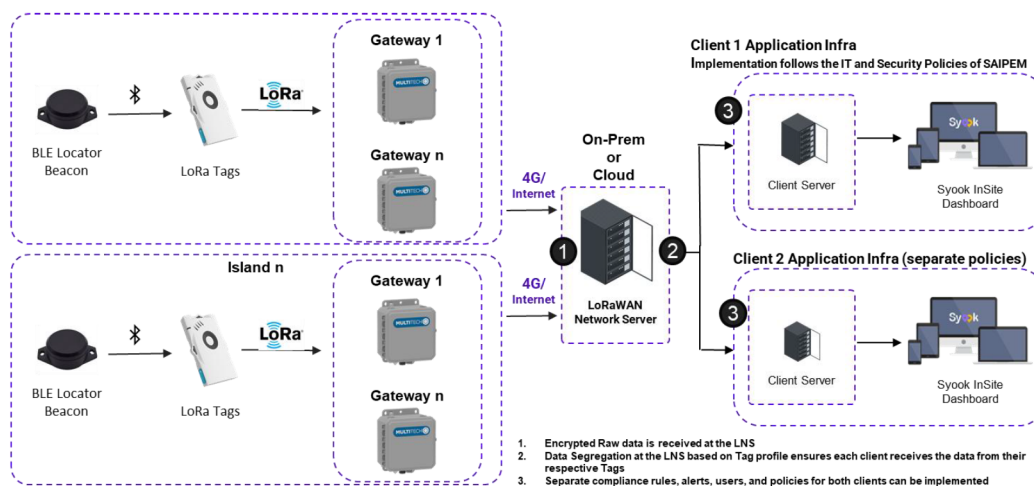


Figure 4—Architecture of e-tracking system for distribution of information

As a reference for utilization, when mustering was performed with the utilization of T-Card system, the minimum time achieved was 47 minutes for head count confirmation. The maximum time for evacuation is expected to be 25 minutes to reach the muster area from the most remote location. In cases where Search and Rescue (S&R) is required, the timing would vary according to the capabilities of the personnel to search, in some cases failing on the search and rescue attempt.

With the utilization of the e-Tracking, the LoRa Tags provide alerts for the status of personnel, thus localization is limited to the accuracy of the LoRa signal, usually within 2m. Thus, ensuring effective recovery measures are taken. E-Mustering is confirmed within seconds and real time onboarding at the muster area can be seen in the control screens. The mustering time from Alarm activation to POB account has not exceeded 23 minutes and consistently achieving 100% of personnel location and account at all drills so far.

Another important aspect is the effect on human factors such as cultural diversity in workforce, language barriers, and psychological readiness during emergencies. Sometimes cultural barriers impede the proper head count, or communication in case of emergencies. This aspect is eliminated when it counts the most, during emergencies. The pressure of evacuation in a stressful situation sometimes makes communication in natural language difficult, in addition the utilization of masks during an emergency is an additional factor to provide guidance to the Tactical Response Team in cases of search and rescue. However, with direct guidance and visual access to the location of interest the command and control of the TRT is simplified and some of those barriers are eliminated. This also impact the moral and psychological health of the personnel that acknowledge the essential aspect of receiving help at all conditions, even when they are not able to call for support.

Access Control System

Access control to the field from the port of embarkation is essential to ensure safe operations, even with the long-range solutions per island that comes with the e-tracking. As the field size extends over 50Km in span and have people in transit or in operation a very different areas within the field, it is essential to keep control of the People on Board vessels during transport, or vessels performing duties. In this way it is also possible to utilize to maximum capacity the available infrastructure and take decisions with regards to the placement of personnel in case of a level/tier 3 type emergency, where the whole field requires evacuation. The access control is based on RFID (Radio Frequency IDentification) which is based on passive access cards that are issued to all authorized personnel to work in the field. In this way it does not only serve the purpose of emergency response, but also the cards hold the status of critical information with regards to medical fitness, required training, security clearances, etc.

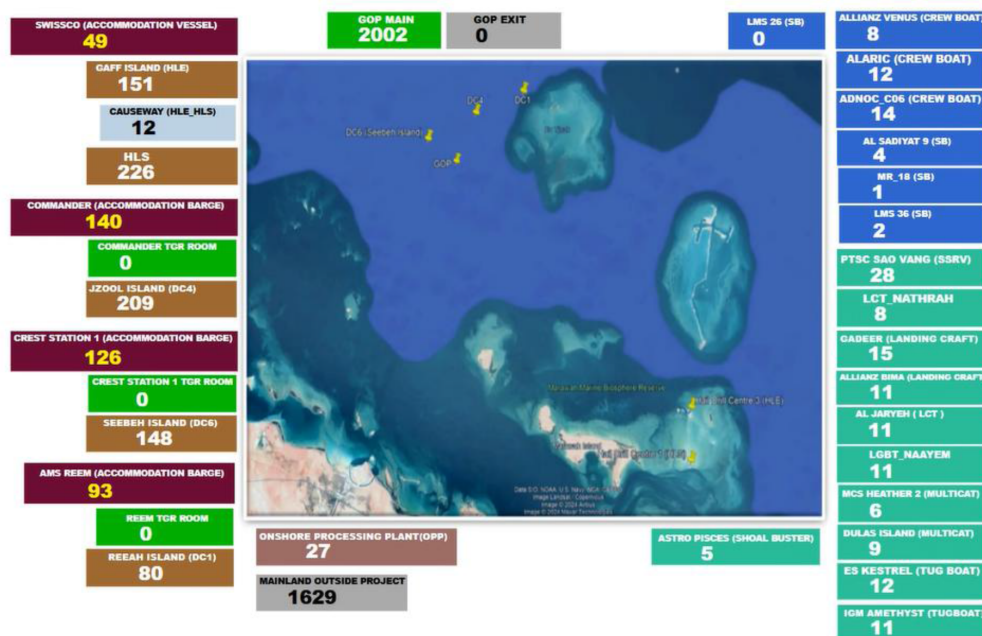


Figure 5—Access control system screen view for all field operations and boat transport

Marine Traffic system

The Marine traffic system is a global vessel tracking platform that uses AIS (Automatic Identification System) data to monitor ship movements in real time. In the UAE, the installation and continuous operation of AIS (Automatic Identification System) is mandatory for all vessels, regardless of size, type, Flag (local or foreign). Here is how it works:

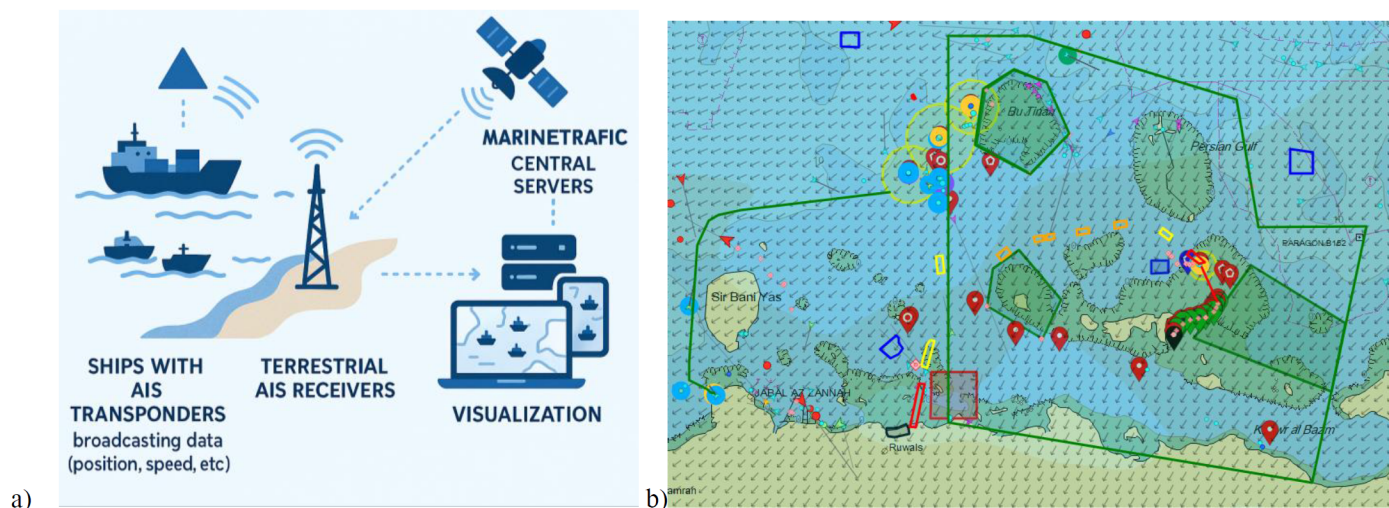


Figure 6—a) The principle of transmission of AIS data from boats to visualization, b) Example of visualization

The application allows the personalization of information on the Geographic Information System, including weather data information such as wind direction, speed, and temperature. These are critical parameters to take the right decision of movement, or preparedness for any condition. As all the vessels in the Hail and Ghasha project are provided with AIS transponders, it is possible to identify the fleet utilized by the project. In addition to tracking vessel movements, estimating arrival times, and accessing historical operational data, the system enables the rapid identification of third-party vessels operating near project

transport routes. In the event of an undesired situation, the Marine Control Center located in the field can identify and contact these vessels using standard marine VHF channels to issue necessary alerts and instructions.

The ability to monitor conditions of operation when an event is ongoing, also allows the possibility to establish a continuously updated register of affected parties. Particularly important are the sensitive receptors in the Marwah Marine Biosphere Area, that has permanently manned rangers in the Butinah Island. Communication and close coordination are established in case of emergency and tested during drills.

Drills and exercises

At peak activity the field will have around 5000 people working in more than 15 locations including Artificial Islands, pipeline laying vessels, and heavy lifting vessels. Thus, to ensure the adequate adaptation and training of personnel the HSE teams have established a stringent training program that intent to have a risk-based approach. This will ensure that areas with higher risk like drilling centers operating in SIMOPS conditions are capable of rapid evacuation of all non-essential personnel, but also the action of Tactical Response Teams in charge of Emergency or Crisis control.

Use Case in Daily Operations

Daily operations also benefit in great manner of these systems. As they provide critical information about the whereabouts of the people in a relatively small and congested area like the artificial islands. The geo-clusters are used to describe areas where personnel should not be present due to the inherent risk of ongoing activities. Therefore, automatic signals can be sent to personnel that inadvertently enter a location that is not authorized. Historical data has also proven beneficial in cases of personnel optimization or as part of incident investigation.

Conclusions and Recommendations

Safety of personnel is paramount to the development of a project. In Hail and Ghasha, where the risk inherent to the required activity to initiate production of the field is a given, the utilization of technology and adequate management of the resources allows the development of activities in a way to ensure that even under emergency conditions, the personnel is safe.

Continuous training in evacuation procedures and toxic gas response is essential, especially for new personnel. The project's Emergency Response Framework recognizes that human factors, such as stress and decision-making under pressure, can significantly influence the outcome of emergency situations.

The application of the Emergency Response Framework tailored for the realities of the project has proven adaptable. The unique risks of this field development project, such as well-control events in ultra-sour reservoirs, and simultaneous construction of facilities with different contractors. Therefore, the separation of TRT and EST structures across contractors reflects the need for tailored responses based on scope and operational risk, while also highlighting the need for collaboration and communication between the different entities.

The implementation of a LoRa-based e-tracking system ensures continuous personnel monitoring across the field. This can be achieved even under extreme emergency conditions such as toxic gas events or power outages. Its low-power, solar-powered architecture with satellite communication backup (VSat) that provides high reliability and operational continuity.

The RFID-based access control system enhances field safety by maintaining real-time visibility of personnel from port embarkation to field deployment. It supports not only emergency response but also operational efficiency by integrating critical data such as medical fitness, training status, and security clearances.

The integration of GIS-based visualization, weather overlays, and real-time personnel tracking empowers the whole Emergency Response Organization (IMT, EST and CMT) to make informed decisions during

emergencies that relate to the need of tactical or strategic response. This enhances the overall capability and safety of operations in a geographically dispersed and high-risk environment.

Finally, the platform is fully aligned with corporate IT infrastructure. Developed entirely within Power BI and integrated securely with SharePoint, the tool maintains data integrity and compliance within the company's security protocols, eliminating reliance on third-party systems.

Acknowledgments

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We also acknowledge the ADNOC Digitalization team for their technical guidance in optimizing dashboard performance and securing integration with corporate infrastructure.

Nomenclature

Term / Symbol Definition

SimOps	Simultaneous Operations – concurrent execution of multiple tasks within the same operational area.
EPC	Engineering, Procurement, Construction – Referred as a type of contracting class that ensure that all the activity required for successful conclusion of the project is in full responsibility of the contracted company.
HSE	Health, Safety, and Environment
RFID	Radio Frequency Identification

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